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CYBER DEFENSE OPERATIONS

SOC Fundamentals

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SECURITY OPERATIONS CENTER (SOC)

A **Security Operations Center** (SOC) is a centralized team or physical command center that continuously **monitors** an organization's digital infrastructure.

Its **primary goal** is to monitor, detect, investigate, and respond to cyber threats in real time.

Why organization need SOC?



Image source: National Institute of Standards and Technology (NIST) via [Wikimedia Commons](#) / Public Domain.

CORE FUNCTIONS OF SOC

Continuous monitoring - SOC **collects** and **observes** security-relevant telemetry

- **Sources** – SIEM, EDR, Firewalls, IDS/IPS, DNS, Proxy servers, authentication system, cloud services, vulnerability scanners, and endpoint agents.

Monitoring is the foundation of SOC work. Without **reliable telemetry**, every other SOC function becomes weaker.

CORE FUNCTIONS OF SOC

Detection is the process of **identifying** suspicious, unauthorized, or anomalous activity within an organization's digital environment.

Common detection **methods**:

- Signature-based alerts
- Behavioral analytics
- Threat intelligence matches
- Manual threat hunting

Detection is **not the same** as confirmation.

CORE FUNCTIONS OF SOC

Alert Triage is the systematic process of reviewing, validating, and prioritizing incoming security alerts to determine whether they represent real threats or false alarms.

- Triage is risk-based decision-making under uncertainty.

Why does it matter?

- Prevent alert fatigue
- Reduce response time
- Prioritize resources

CORE FUNCTIONS OF SOC

Investigation and **event correlation**: Once an alert appears suspicious, the analyst must **gather** evidence, **correlate** events, and **reconstruct** what happened.

- Investigation usually involves pivoting across data sources.
- SOC team performs **detailed investigations** to understand the root causes, methods, and impact of the attack

Threat intelligence helps the SOC understand adversaries, campaigns, malware, infrastructure, tactics, techniques, and procedures.

- Low-quality IOC feeds can create noise.

CORE FUNCTIONS OF SOC

Escalation occurs when an alert or investigation suggests that **action** is needed beyond routine triage.

- The SOC may escalate to Tier 2 analysts, incident responders, threat hunters, malware analysts, system owners, cloud administrators, network teams, legal/compliance, or management.

Threat hunting is structured, hypothesis-driven investigation designed to find threats that automated detections may have missed.

When a security incident is identified, the SOC team initiates an **incident response** process.

- This process involves containment, eradication, recovery, and post-incident activities.
- Relation between SOC and CSIRT (Computer Security Incident Response Team)

CORE FUNCTIONS OF SOC

Regular **reporting** is an essential SOC process to provide insights into security incidents, trends, and vulnerabilities.

- The SOC team prepares reports that summarize the activities, findings, and recommendations.

Continuous improvement

- Every major or minor security incident provides a **roadmap** for better defense.
- Requires a constant **feedback** loop between the analysts who handle alerts and the detection engineers who build them.
- As analysts identify **repetitive** manual tasks during investigations, the continuous improvement team converts those actions into automated code.
- **Vulnerability management**: Identify and address vulnerabilities
- Security **awareness** training

SOC MODELS

Internal SOC

- Owned and operated by the organization it serves
- Monitors and defends internal networks, systems, and data

Managed SOC (MSOC)

- Third-party provider of security operations
- Basic monitoring or comprehensive threat detection and response

Hybrid SOC

- Elements of internal and managed SOC services
- Flexible compromise - call in the experts as needed

METRICS FOR EVALUATING SOC EFFECTIVENESS

MTTD (Mean Time to Detect): Average time from malicious activity to detection

MTTR – (Mean Time to Respond or Remediate): Average time from detection to containment/remediation

False Positive Rate (**FPR**): Percentage of alerts that turn out to be benign.

False Negative Rate (**FNR**): Percentage of actual attacks that go unnoticed by the security stack.

Detection Coverage: The percentage of threat techniques (such as those mapped to the MITRE ATT&CK framework) that your SOC is actively instrumented to catch.

Mean Time to Attend and Analyze (**MTTA&A**): measure how fast analysts triage and investigate alerts before taking action.

SOC TOOL ECOSYSTEM

Security Information and Event Management (SIEM)

Endpoint Detection and Response (EDR)

Intrusion Detection and Prevention Systems (IDS/IPS)

Security Orchestration, Automation, and Response (SOAR)

Network Security Monitoring (NSM)

Threat Intelligence Platforms (TIP)

Forensic Analysis Tools

Malware Analysis Tools

Firewalls

- Network Firewalls
- Next-Generation Firewalls (NGFW)
- Web Application Firewall (WAF)

SOC DATA AND TOOLS

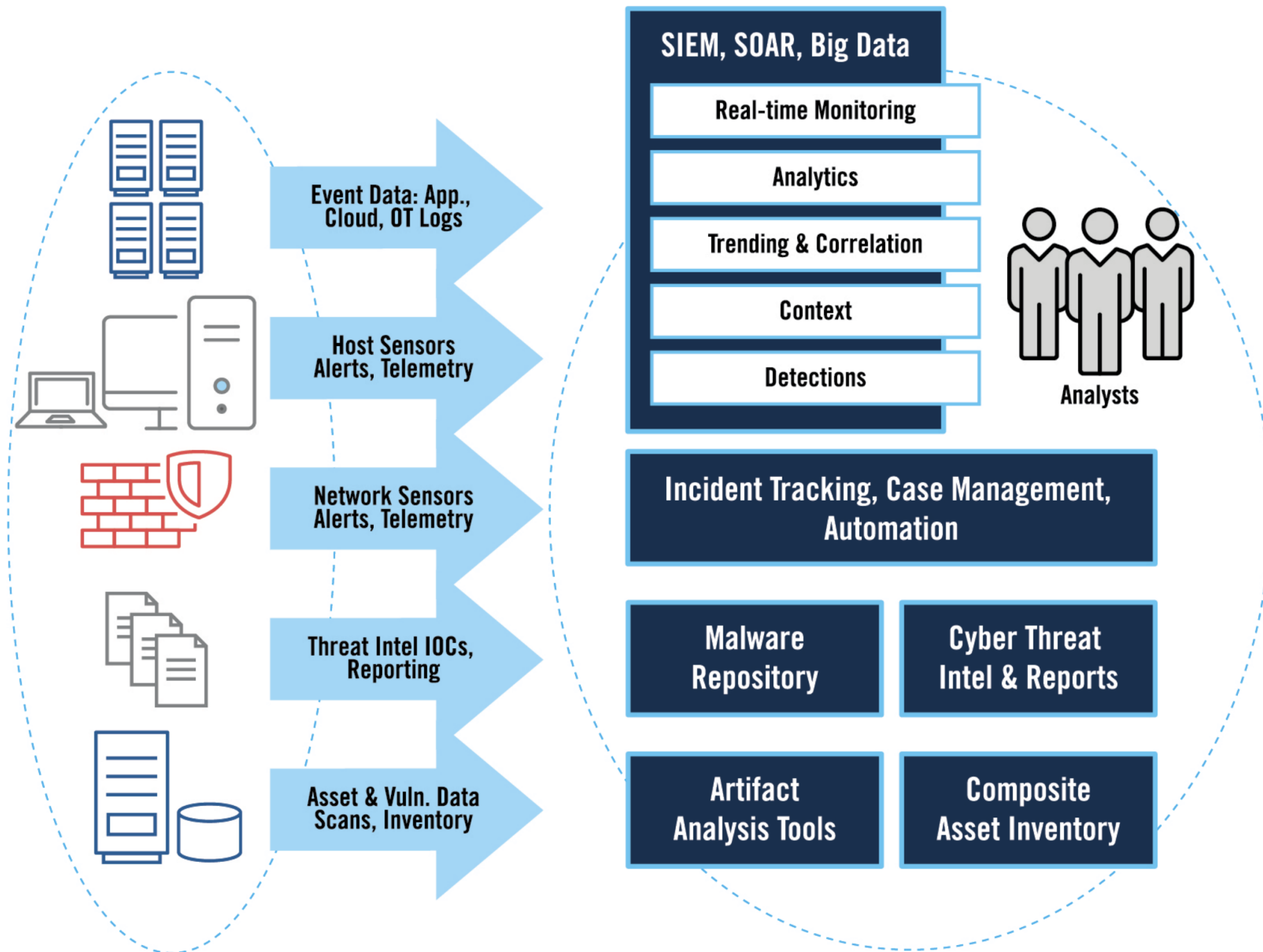


Figure 1. Source: Knerler, K., Parker, I., & Zimmerman, C. (2022). 11 Strategies of a World-Class Cybersecurity Operations Center. [The MITRE Corporation](#).

SOC ROLES

SOC Role	Brief Definition
Tier 1 Analyst	Performs initial alert monitoring and triage . Reviews incoming alerts, filters false positives, gathers basic evidence, and escalates suspicious cases.
Tier 2 Analyst	Conducts deeper investigation of escalated alerts. Correlates logs, analyzes suspicious activity, scopes incidents, and determines whether response action is needed.
Tier 3 Analyst	Handles advanced investigations and complex threats . Performs threat hunting, malware behavior analysis, advanced correlation, and detection improvement.
SOC Team Lead	Supervises SOC analysts and daily operations. Reviews escalations, manages shift handovers, ensures process quality, and supports analyst development.
Incident Responder	Coordinates and executes incident handling activities , including containment, eradication, recovery, evidence preservation, and post-incident review.
Threat Hunter	Proactively searches for hidden threats that may not have triggered alerts. Develops hypotheses, analyzes anomalies, and identifies visibility gaps.
Threat Intel Analyst	Collects, analyzes, and applies threat intelligence . Provides context on threat actors, campaigns, indicators of compromise, and adversary tactics.

SOC ROLES

SOC Role	Brief Definition
Security Engineer	Builds, maintains, and improves SOC tools and security infrastructure, such as SIEM, EDR, IDS, logging pipelines, and detection integrations.
Vulnerability Manager	Identifies, prioritizes, and tracks vulnerabilities across systems. Works with IT and security teams to reduce exposure and support remediation.
Forensic Analyst	Performs detailed forensic examination of systems, files, memory, logs, and disk images to determine what happened during an incident.
Malware Analyst	Analyzes malicious software to understand behavior, capabilities, persistence, indicators, and potential impact. Supports detection and response.
SOC Manager	Manages SOC operations, staffing, processes, performance metrics, reporting, and coordination with other security and IT teams.
Director of Security	Oversees broader security operations and strategy. Aligns SOC activities with organizational risk, compliance, budget, and security objectives.
CISO	Senior executive responsible for the organization's overall cybersecurity strategy, governance, risk management, security policy, and executive reporting.

REFERENCES

[1] Knerler, K., Parker, I., & Zimmerman, C. (2022). 11 Strategies of a World-Class Cybersecurity Operations Center (2nd ed.).

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[2] Murdoch, D. (2024). Blue team handbook: SOC, SIEM, and threat hunting: Practical techniques for security operations and threat hunting teams. O'Reilly Media.

Questions?